

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:		Duration:	60 minutes

Code of Conduct

I K Y V Satyanarayana reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *K Y V Satyanarayana reddy*

Annexure A – Scenario-Based Requirement Analysis

Business Intelligence Dashboard for Executive Decision Support

1. Study and Analyse the Scenario

1.1 Understanding the Problem Statement

Organizations generate large volumes of business data from departments such as sales, finance, marketing, inventory, and human resources. Most organizations face challenges in converting this raw data into meaningful information for decision-making. Managers often rely on multiple reports from different systems, making it difficult to obtain accurate and real-time insights. The proposed **Business Intelligence Dashboard for Executive Decision Support** integrates data from multiple sources, analyzes it, and presents it through interactive dashboards, charts, and reports. This enables executives to monitor organizational performance, identify trends, and make informed strategic decisions quickly and efficiently.

1.2 Business Domain

The project belongs to the **Business Intelligence and Data Analytics** domain. It focuses on collecting, processing, analysing, and visualizing organizational data to support executive-level decision-making. The system converts raw business data into meaningful reports and dashboards that improve planning, forecasting, and performance evaluation across different departments.

1.3 Stakeholders

Primary Stakeholders

- Executive Management (CEO, CFO, COO)
- Business Managers
- Department Heads
- Data Analysts
- System Administrator

Secondary Stakeholders

- Employees
- Customers
- Investors
- IT Support Team
- Regulatory Authorities

Roles and Responsibilities

- Executives monitor overall business performance.
- Managers analyze departmental performance.
- Data Analysts prepare reports and dashboards.
- Administrators manage users and system security.
- IT Team maintains the application and database.
- Investors review business growth reports.
- Employees update operational data.
- Regulatory authorities verify compliance reports.

1.4 Existing Challenges

- Business data is stored in multiple systems.
- Reports are prepared manually.
- Delay in generating business reports.
- Difficulty in identifying business trends.
- Lack of real-time data analysis.
- High possibility of human errors.
- Poor decision-making due to outdated information.
- Limited graphical visualization of business performance.
- Data inconsistency across departments.
- Inefficient communication between departments.

1.5 System Objectives

- Provide a centralized business dashboard.
- Display real-time business information.
- Generate accurate reports automatically.
- Improve executive decision-making.
- Reduce manual reporting effort.
- Enhance data visualization.
- Improve operational efficiency.
- Support business forecasting.
- Ensure secure data access.
- Increase overall organizational productivity.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

1. User Login and Authentication
 2. User Role Management
 3. Dashboard Creation
 4. Data Import from Multiple Sources
 5. Real-Time Data Visualization
 6. Sales Performance Analysis
 7. Financial Report Generation
 8. Inventory Analysis
 9. Employee Performance Monitoring
 10. Customer Analytics
 11. KPI Monitoring
 12. Report Generation
 13. Export Reports to PDF and Excel
 14. Notification and Alert System
 15. Search and Filter Reports
 16. Data Backup and Recovery
 17. Audit Log Management
 18. User Profile Management
 19. Dashboard Customization
 20. Data Refresh Functionality
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2.2 Non-Functional Requirements

Performance

- Dashboard should load within 3 seconds.
- Reports should generate quickly.
- Support large datasets.

Security

- Secure login authentication.
- Role-based authorization.

- Data encryption.
- Secure database access.

Reliability

- Continuous system availability.
- Accurate data processing.
- Automatic backup.

Usability

- User-friendly interface.
- Easy navigation.
- Interactive dashboards.

Scalability

- Support increasing users.
- Handle growing business data.
- Easy addition of new modules.

Maintainability

- Easy software updates.
- Modular architecture.
- Simple debugging process.

Availability

- System available 24×7.
- Minimum downtime.
- Disaster recovery support.

3. Identify Actors and Use Cases

3.1 Primary Actors

- Executive
- Business Manager
- Data Analyst
- Administrator

3.2 Secondary Actors

- Employee
 - Database Server
 - External Data Source
 - Reporting System
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3.3 Major Use Cases

Executive

- View Dashboard
- Analyse KPIs
- Generate Reports
- View Business Trends

Manager

- Monitor Department Performance
- Analyse Sales
- View Inventory
- Download Reports

Data Analyst

- Upload Data
- Clean Data

- Generate Charts
- Create Reports

Administrator

- Manage Users
 - Assign Roles
 - Backup Database
 - Configure Dashboard
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3.4 Use Case Diagram

4. Software Requirement Specification (SRS)

4.1 Introduction

The Business Intelligence Dashboard for Executive Decision Support is designed to help organizations transform raw business data into meaningful information. The system provides interactive dashboards, charts, and reports that assist executives in monitoring business performance and making strategic decisions. It integrates data from multiple departments and presents it in an easy-to-understand visual format.

4.2 Problem Description

Organizations often struggle with scattered data, delayed reporting, and manual analysis. Decision-makers spend significant time collecting information from multiple systems before making strategic decisions. The proposed Business Intelligence Dashboard solves these problems by integrating business data into a centralized platform that provides real-time visualization, automated reporting, and analytical insights.

4.3 Scope of the System

The system supports business performance monitoring, financial analysis, sales analysis, inventory tracking, employee performance evaluation, customer analytics, and report generation. It provides executives with real-time dashboards that improve planning, forecasting, and strategic decision-making. The system can be implemented in organizations of various sizes and can be expanded with additional business modules in the future.

4.4 Functional Requirements

(Use the Functional Requirements listed in Section 2.1.)

4.5 Non-Functional Requirements

(Use the Non-Functional Requirements listed in Section 2.2.)

4.6 Use Case Descriptions

Use Case	Description
Login	User logs into the dashboard securely.
View Dashboard	Displays KPIs and business metrics.
Upload Data	Imports business data into the system.
Generate Report	Creates business reports.
Analyse Trends	Displays historical business trends.
Export Report	Saves reports as PDF or Excel.

4.7 Data Requirements

- Employee Data
 - Sales Data
 - Customer Data
 - Financial Data
 - Inventory Data
 - Supplier Data
 - Product Information
 - User Login Information
 - KPI Data
 - Business Reports
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4.8 External Interface Requirements

User Interface

- Graphical dashboard
- Responsive web interface
- Interactive charts

Hardware Interface

- Computer
- Server
- Internet connection

Software Interface

- Database Management System
- Web Browser
- Operating System

Communication Interface

- Internet
 - Local Network
 - Cloud Services
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4.9 System Features

- Interactive Dashboard
 - Real-Time Data Visualization
 - Business Analytics
 - KPI Monitoring
 - Automatic Report Generation
 - Role-Based Access Control
 - Secure Login
 - Data Filtering
 - Dashboard Customization
 - Export Reports
 - Data Backup
 - Notification System
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5. Assumptions and Constraints

5.1 Assumptions

- Users have basic computer knowledge.
- Internet connectivity is available.
- Business data is accurate.
- Authorized users access the system.
- Database server functions properly.
- Management supports system implementation.

- Required hardware is available.
 - Data sources remain accessible.
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5.2 Constraints

Technical Constraints

- Limited server resources.
- Database compatibility issues.
- Internet dependency.

Operational Constraints

- User training required.
- Data migration challenges.

Economic Constraints

- Software licensing costs.
- Hardware expenses.
- Maintenance costs.

Legal Constraints

- Data privacy regulations.
- Compliance with organizational policies.

Time Constraints

- Limited project duration.
 - Testing deadlines.
 - Deployment schedule.
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6. Requirement Completeness and Consistency

6.1 Validation

The proposed requirements address the needs of all identified stakeholders, including executives, managers, analysts, administrators, and employees. Functional and non-functional requirements are clearly defined and aligned with the project objectives. The requirements are consistent, feasible, and support efficient business intelligence operations. Proper validation ensures that the system meets organizational goals without ambiguity or missing functionality.

6.2 Requirement Validation Checklist

- ✓ All stakeholder requirements are identified.
 - ✓ Functional requirements are complete.
 - ✓ non-functional requirements are clearly defined.
 - ✓ No duplicate requirements are present.
 - ✓ Requirements are consistent.
 - ✓ Requirements are realistic and achievable.
 - ✓ Security requirements are included.
 - ✓ Performance requirements are specified.
 - ✓ User roles are clearly defined.
 - ✓ Business objectives are fully addressed.
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6.3 Suggested Improvements

- Integrate Artificial Intelligence for predictive analytics.
- Add Machine Learning for demand forecasting.
- Support mobile dashboard access.

- Enable voice-based dashboard interaction.
- Integrate cloud storage services.
- Provide real-time notification alerts.
- Include advanced data visualization techniques.
- Support multilingual interfaces.
- Add chatbot assistance for report generation.
- Implement blockchain-based data security.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and	Clearly identifies realistic assumptions and constraints;	Identifies most assumptions and constraints;	Limited assumptions/constraints;	Assumptions, constraints, and validation are

Requirement Validation (10%)	validates requirements for completeness, consistency, and ambiguity with appropriate justification.	validation is adequate.	validation is superficial.	missing or incorrect.
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Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25